



**THE UNIVERSITY
OF QUEENSLAND**
AUSTRALIA

This exam paper must not be removed from the venue

Venue _____
 Seat Number _____
 Student Number

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 Family Name _____
 First Name _____

School of Mathematics & Physics
Semester Two Examinations, 2023
MATH1061 Discrete Mathematics

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

- This is a Closed Book examination - no written materials permitted
- No calculators permitted
- During Planning Time - Students are encouraged to review and plan responses to the exam questions
- This examination paper will be released to the Library

Materials Permitted in the Exam Venue:

(No electronic aids are permitted e.g. laptops, phones)

None

Materials to be supplied to Students:

Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

None

Instructions to Students:

If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

Answer all questions.

Total 70 marks.

The Formula Sheet is on the last page

For Examiner Use Only

Question Mark

Total _____

1. (6 marks)

- (a) (2 marks) Write down the quotient and remainder when -27 is divided by 6 , according to the quotient-remainder theorem.

- (b) (2 marks) Determine whether the following statement is true or false. Justify your reasoning.

For all positive integers a and b , we have

$$\gcd(a + b, a - b) = \gcd(a, b).$$

- (c) (2 marks) Compute $\gcd(112, 30)$. Show your work.

2. (3 marks) Let p and q be statement variables. Are the following statements logically equivalent? Justify your answer.

Statement A: $(p \rightarrow q) \wedge \sim q$

Statement B: $\sim (p \wedge q)$

3. (8 marks)

- (a) (3 marks) Let n be a positive integer and let K_n be the complete graph with n vertices. How many edges does K_n have? Explain your reasoning.

- (b) (5 marks) Let G be a graph. Let y be the number of vertices of G that have odd degree. Prove that y is even.

4. (6 marks) Let $f : \mathbb{N} \rightarrow \mathbb{R}$ be a function. Prove that the image of $\mathbb{N}$ under f is countable.

5. (8 marks) Let $f : \mathbb{Z} \rightarrow \{0, 1, 2, 3, 4\}$ be defined by: $f(n) = n^2 \pmod{5}$.

For example, $f(7) = 4$ because $7^2 = 49$ is congruent to 4 modulo 5.

(a) (2 marks) What is $f(2023)$? Justify your answer.

(b) (2 marks) What is the preimage of 1? Justify your answer

(c) (2 marks) Is f one to one? Justify your answer.

(d) (2 marks) Is f onto? Justify your answer.

6. (8 marks) Let $(G, *)$ be a group. Define a relation $\sim$ on G as follows: $a \sim b$ if there exists $g \in G$ such that $g * a * g^{-1} = b$. Prove that $\sim$ is an equivalence relation on G .

7. (8 marks) Recall that for a set A , its power set is denoted $\mathcal{P}(A)$.

(a) (4 marks) Compute $\mathcal{P}(\{x\})$ and $\mathcal{P}(\mathcal{P}(\{x\}))$.

(b) (4 marks) Prove that

$$\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B).$$

8. (6 marks) Consider the sequence $(a_n)_{n=1}^{\infty}$ defined recursively by $a_1 = 3$, $a_2 = 8$, and

$$a_n = 4(a_{n-1} - a_{n-2}), \quad \forall n \geq 3.$$

Using strong induction, prove that $a_n = 2^{n-1}(n+2)$ for all integers $n \geq 1$.

9. (7 marks) A group of 10 friends are attending the movies together and insist on sitting in one row. There are exactly ten seats in each row.
- (a) (2 marks) Two of the friends (Bailey and Courtney) insist on sitting next to each other. In how many different ways can the 10 be seated while satisfying this wish? Justify your answer.
- (b) (2 marks) Now suppose Bailey and Courtney no longer insist on sitting next to each other. However, there are two other people, Tara and Adam, who do not like each other and do not want to sit next to each other. In how many different ways can the 10 be seated while satisfying this wish? Justify your answer.
- (c) (3 marks) Finally, suppose Bailey and Courtney want to sit next to each other and, at the same time, Tara and Adam do not want to sit next to each other. In how many different ways can the 10 be seated while satisfying these wishes? Justify your answer.

10. (10 marks) Define a binary operation $\circ$ on $G = \mathbb{R}^2 - \{(0, 0)\}$ by

$$(a, b) \circ (c, d) = (ac - bd, ad + bc).$$

(a) (6 marks) Prove that $(G, \circ)$ is an abelian group. (You do not need to prove that the operation is closed or associative.)

(b) (4 marks) Determine all subgroups of $(G, \circ)$ which have exactly two elements. Justify your answer.

End of Examination

MATH1061/MATH7861 Examination Formula Page

Logical Equivalences

Given any statement variables p , q and r , a tautology $\mathbf{t}$ and a contradiction $\mathbf{c}$, the following logical equivalences hold.

<i>Commutative laws</i>	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
<i>Associative laws</i>	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
<i>Distributive laws</i>	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
<i>Identity laws</i>	$p \wedge \mathbf{t} \equiv p$	$p \vee \mathbf{c} \equiv p$
<i>Negation laws</i>	$p \vee \sim p \equiv \mathbf{t}$	$p \wedge \sim p \equiv \mathbf{c}$
<i>Double negative laws</i>	$\sim(\sim p) \equiv p$	
<i>Idempotent laws</i>	$p \wedge p \equiv p$	$p \vee p \equiv p$
<i>Universal bound laws</i>	$p \vee \mathbf{t} \equiv \mathbf{t}$	$p \wedge \mathbf{c} \equiv \mathbf{c}$
<i>De Morgan's laws</i>	$\sim(p \wedge q) \equiv \sim p \vee \sim q$	$\sim(p \vee q) \equiv \sim p \wedge \sim q$
<i>Absorption laws</i>	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
<i>Negations of $\mathbf{t}$ and $\mathbf{c}$</i>	$\sim \mathbf{t} \equiv \mathbf{c}$	$\sim \mathbf{c} \equiv \mathbf{t}$

Set Identities For all sets A , B and C that are subsets of a universal set U :

1. Commutative Laws (a) $A \cup B = B \cup A$ (b) $A \cap B = B \cap A$
2. Associative Laws (a) $(A \cup B) \cup C = A \cup (B \cup C)$
(b) $(A \cap B) \cap C = A \cap (B \cap C)$
3. Distributive Laws (a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4. Identity Laws (a) $A \cup \emptyset = A$ (b) $A \cap U = A$
5. Complement Laws (a) $A \cup A^c = U$ (b) $A \cap A^c = \emptyset$
6. Double Complement Law $(A^c)^c = A$
7. Idempotent Laws (a) $A \cup A = A$ (b) $A \cap A = A$
8. Universal Bound Laws (a) $A \cup U = U$ (b) $A \cap \emptyset = \emptyset$
9. De Morgan's Laws (a) $(A \cup B)^c = A^c \cap B^c$ (b) $(A \cap B)^c = A^c \cup B^c$
10. Absorption Laws (a) $A \cup (A \cap B) = A$ (b) $A \cap (A \cup B) = A$
11. Complement of U and $\emptyset$ (a) $U^c = \emptyset$ (b) $\emptyset^c = U$
12. Set Difference Law $A - B = A \cap B^c$

The Quotient-Remainder Theorem Given any integer n and any positive integer d , there exist unique integers q and r such that $n = dq + r$ and $0 \leq r < d$.

Unique Factorisation Theorem Given any integer $n > 1$, there exists a positive integer k , distinct prime numbers $p_1, p_2, \dots, p_k$, and positive integers $e_1, e_2, \dots, e_k$ such that $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$, and any other expression for n as a product of prime numbers is identical to this except perhaps for the order in which the factors are written.

Schröder-Bernstein Theorem For all sets A and B , if $|A| \leq |B|$ and $|A| \geq |B|$ then $|A| = |B|$.

Binomial Theorem Given any real numbers a and b , and any nonnegative integer n ,

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k.$$